PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

MUSIC PLAYLIST MANAGER

Student / Team Information

Team Name: *Team DSCPP-III-2026-T326*	*dictator*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Rawat , Tanishq – 2510012727 tanishqrawat201@gmail.com* 
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Music has become a common part of everyday life, and people usually listen to different songs based on their mood activity or simply their own choice.A simple playlist management system can help make these tasks easier and keep everything organized.

A Queue is used to handle the upcoming songs where songs are played in the same order in which they were added. This makes it easier to manage the songs that are waiting to be played.

by developing a music playlist manager we aim to connect the theoretical concepts of dsa to a familiar real world application hence this also gives a better idea of different data structures can be also used together to solve a practical problem in a useful way.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Today popular music streaming applications provide advanced playlist management features.Users can create playlists and can add or remove songs also search for music play songs in different orders and view their listening history.

The playlist is managed using a linked list recently played songs are stored using a stack and upcoming songs are handled using a queue.

It simulates the working of a music player through song names and artist information throughout and it is to understand how data can be organized and could be stored and processed efficiently using different data structures.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The main goal of this project is to develop a basic Music Playlist Manager using C++ and fundamental data structures.The project will demonstrate the practical implementation of linked lists ,stacks,queues in a single application.

The first goal is to create a main playlist using a linked list so the users can add songs, delete songs, search for songs, and display the complete playlist.

The second goal is to implement a stack for managing recently played songs and when a song is played it will be added to the stack.

The third goal is to implement a queue for managing upcoming songs as the songs would be added to the queue and be played according to the fifo principle as the first song added would be played first.
The initial milestones of the project are:

1. *Understanding the required Data Structures and project requirements.*
2. *Designing the Song node and Linked List structure.*
3. *Implementing playlist operations such as add,delete, search,display.*
4. *Implementing the Recently Played feature using Stack.*
5. *Implementing the Upcoming Songs feature using Queue.*
6. *Creating a menu-driven C++ program.*
7. *Debugging errors and improving the overall program.*
8. *Preparing the final project report and presentation.*

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max
300 words).

It is a console based application will be created because the main purpose of the project is to understand data structures rather than focus on graphical user interface development.

The linked list will provide operations like adding a song,deleting a song,searching for a song,and displaying the complete playlist.

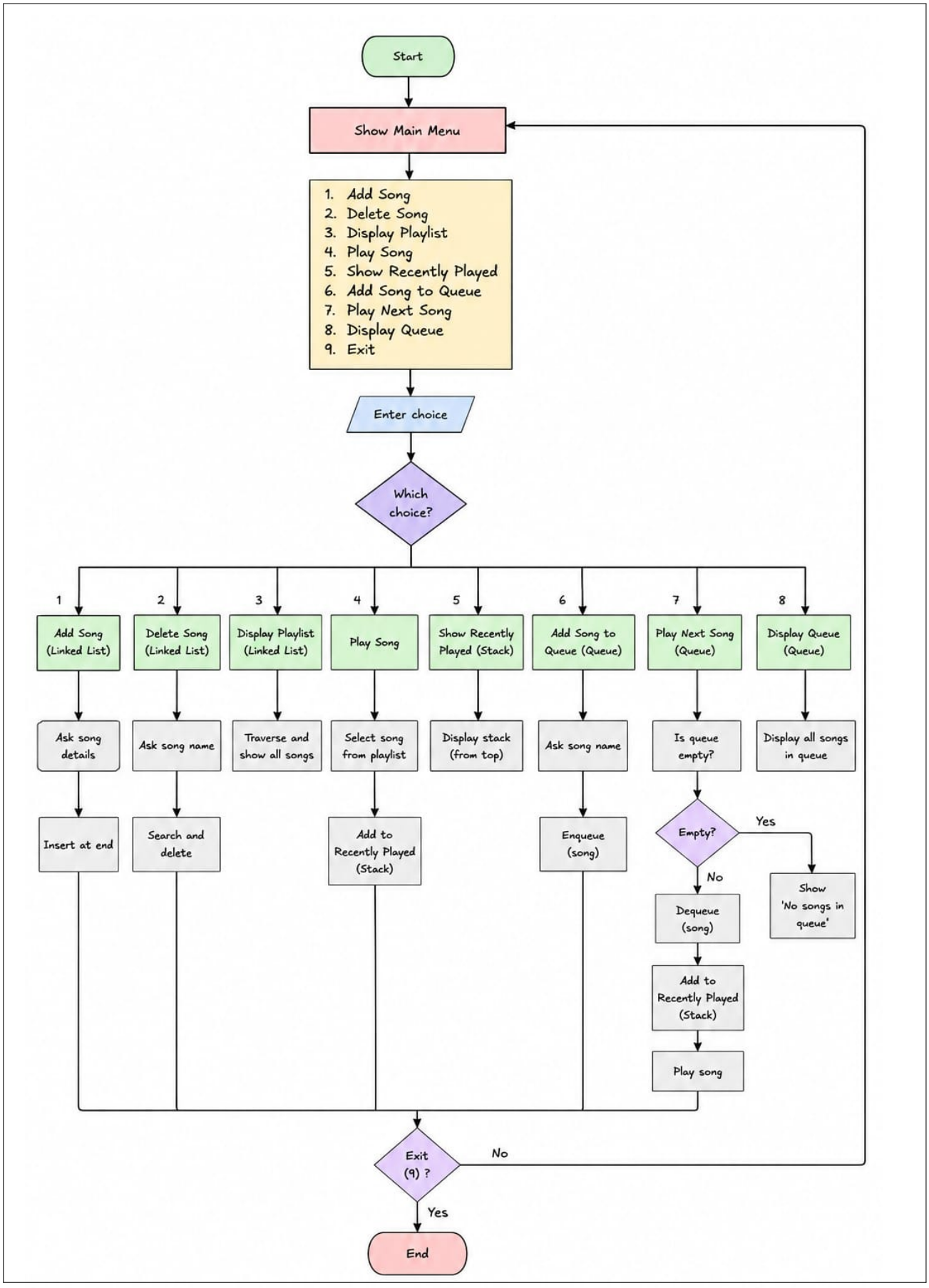
stack could be implemented to manage recently played songs also the user can then view the recently played songs based on the last in first out principle.

Queue would be implemented for upcoming songs and the users can add songs to the play queue and songs will be played according to the first in first out principle.

The program will use a menu-driven interface.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The application will allow the users to perform basic playlist operations such as adding a song also deleting a song moreover searching for a song and displaying all available songs done by using a linked list.

The project will include a recently played feature used by stack so it will be stored in the stack so that the user can view the recently played songs.

An upcoming songs feature will also be included using a queue so that the users would be able to add songs to the queue and can also play them as their order in which they are added.

The final deliverables of the project will include:

1. *Complete C++ source code*
2. *A working console based application*
3. *Implementation of linked list,stack and queue*
4. *Project documentation and report*
5. *System architecture diagram*
6. *Screenshots of the program execution*
7. *A presentation explaining the project and Data Structures used.*

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

This project assumes that the application is used by a single user at a time.

The songs would be represented only by basic information like song name and artist name and the application doesn't actually stream or play audio files.

It is assumed that user enters valid information while adding songs the project focuses mainly on demonstrating data structures and basic playlist management operations.

The program will run locally on a computer using a C++ compiler.Internet connectivity,databases,user accounts,and online music streaming services are not required for the basic working of the system.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. *C++ Programming Documentation and Standard Library references.*
2. *GeeksforGeeks – Data Structures and Algorithms https://www.geeksforgeeks.org/data-structures/*
3. *TutorialsPoint – Data Structures Tutorials https://www.tutorialspoint.com/data$_s$tructures$_a$lgorithms/*
4. *cppreference – C++ Language and Standard Library Reference https://en.cppreference.com/*
5. *Course notes and classroom material related to Linked Lists, Stacks, and Queues.*